FYSC22

Gamma spectroscopy

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September 15, 2024

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1 Introduction

An atom can emit light by x-rays which are the de-excitations of the electrons or gamma rays which come from the de-excitation of the nucleus. In this lab, the gamma emission for different nuclei was evaluated. This was done through gamma spectroscopy where samples were placed near detectors covered in lead shielding. This form of spectroscopy was in the lab used to identify different types of material and their corresponding spectra.

2 Theory

2.1 Gamma ray interaction

There are different ways a gamma ray can interact depending on its energy. One such way is via the photoelectric effect, which can occur with very low energy photons as seen in Figure 1a. The photoelectric effect occurs by an electron absorbs an incoming gamma ray which in turn gets knocked. This process can be described with a certain binding energy ϕ as

$$E_e = E_\gamma - \phi. \tag{1}$$

If the gamma ray has a little more energy it can Compton scatter an electron. In Compton scattering is not the entire gamma ray absorbed by the electron. This leads to scattering between the electron and the photon with a certain angle θ. Compton scattering can be described as

$$E'_\gamma = \frac{E_\gamma}{1 + \frac{E_\gamma}{m_e c^2}\left(1 - \cos(\theta)\right)}, \tag{2}$$

where E'_γ is the energy of the scattered gamma ray, E_γ is the energy of the incoming gamma ray and m_e is the mass of the electron.

If the energy of the gamma ray is larger than two electron masses m_e there is an possibility of pair production. Pair production can be described as the energy of the gamma ray converts into an electron positron pair as

$$T_{e^-} + T_{e^+} = E_\gamma - 2m_e c^2, \tag{3}$$

where T_e is the kinetic energy of the particles. For the detector can this process be described as

$$E_{\text{det}} = T_{e^-} + T_{e^+} + 2T_{e^-_\gamma}. \tag{4}$$

Another form of emission that can occur for excited nuclei is internal conversion. This process transfers the energy into one of the electrons in the atom resulting in electron emission. These electrons are called auger electrons. The internal conversion coefficient can be described as

$$\alpha = \frac{I_0^{\text{x-ray}}}{I_0^\gamma} \tag{5}$$

where $I_0^{\text{x-ray}}$ is the emitted x-rays and I_0^γ is the emitted gamma rays from a certain transition. However, due to the auger electrons which compete with this process, we must correct this term with the fluorescence yield ω. The corrected internal coefficient can then be calculated as

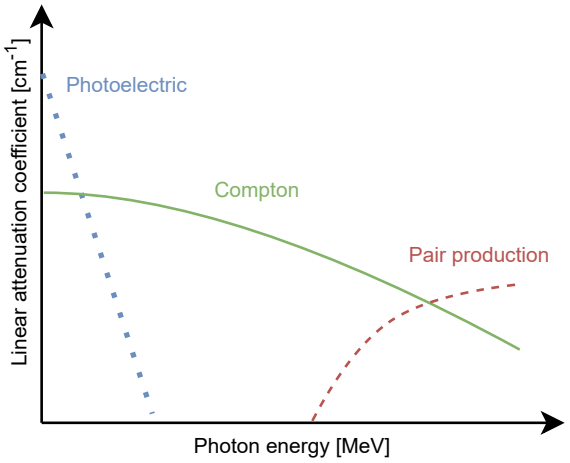
$$\alpha_{\text{corrected}} = \frac{\alpha}{\omega}. \tag{6}$$

2.2 Gamma ray detection

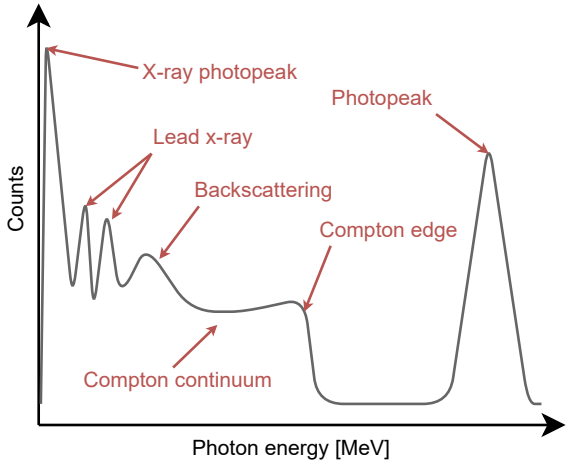
A Scintillator detector works by having some scintillator material absorbing incident radiation and exciting multiple electrons in the material. These excited states then decay and send out many lower-energy photons which are then picked up by the photomultiplier tube (PMT). The scintillator material has many closely spaced energy levels as well. When the photon enters the PMT it first reacts with a photocathode which converts the photon to an electron via the photoelectric effect. The electrons are the focused and accelerated to a dynode with an electric field. When the electron hits the dynode it releases around 5 times more electrons, which are then accelerated towards another dynode. This continues between 10-14 dynodes and is picked up by an anode. Common types of scintillators in gamma spectroscopy is thallium-doped sodium iodide (NaI(Tl)) or bismuth germanate (BGO).

Semiconductors are another way of detecting gamma rays. These detectors work by the use of two sorts of materials n-type and p-type. The n-type material has electrons which easily are excited into a conduction band, while the p-type has electrons who are easily deexited creating holes. Thus when a n-type is placed near a p-type together with an electric field a p-n junction is created. The energy gap between the valence band and the conduction band in the p-n junction is small, so when a gamma ray hit this surface is an electron easily promoted to the conduction band where the electric field moves the electron to the p-side and the hole to the n-side. Common types of semiconductors for gamma spectroscopy is germanium detectors which has a significantly better resolution than the scintilator. However a disadvantage with this type of detectors is that they are very thermal sensitive meaning that they need to be kept at cryogenic temperatures to work properly.

Compton scattering and photo electrons can be detected by a scintillator or a semiconductor which can look like Figure 1b. There one can see that a large amount of gamma rays are detected through the photoelectric effect in what is labelled as the photopeak. One can also see that some energy is deposited by Compton scattering, where the Compton edge shows Compton scattering with 180°. The label backscattering shows gamma rays which are detected after being scattered by the lead shielding before being detected. The peaks displaying "lead x-rays" come from the x-ray emission of the lead shielding whom are excited by gamma rays. The other peak labelled x-ray photopeak arises from the internal conversion. Annihilation peaks is another relevant phenomena which is not displayed. These peaks come from the annihilation of the positron (β^+) resulting in the emission of a gamma ray with the same energy as that of the rest mass of the electron/ positron. This would also give peaks in Figure 1b.



(a) This figure shows how gamma rays interact depending on it's energy.



(b) This figure shows how the expected graph of gamma spectroscopy should look like.

Figure 1: In the two figures above can one see how the different energies on the photon affect our measurement. The drawings were inspired by [1].

If we want to detect high-energy gamma rays can see in Figure 1a that this is mostly done through pair production. This can be done with a BGO since it has a high density and high atomic numbers which means it can detect high energy gamma rays [2]. This plot would show different escape peaks which would be separated from the main peak with an energy difference of the rest mass of an electron. This is due to the fact that the scintillator might not detect one or two of the produced gamma rays from the produced electron/ positron pair. Thus the detector sometimes detects one or two electron masses too little, known as escape peaks.

Due to the fact that the gamma emission is isotropic and the fact that the detectors do not detect all the emission must a formula be used to find the total emission. The total emission from a source can be described as

$$I_0 = \frac{4\pi}{\Omega} \frac{I_{\text{fe}}}{\epsilon_{\text{eff}} \cdot \epsilon_{P/T}}, \tag{7}$$

where $\Omega/4\pi$ is the angular coverage, I_{fe} is the gamma-rays leaving full energy in the detector, ϵ_{eff} is the efficiency of the detector and $\epsilon_{P/T}$ is the peak-to-total-ratio.

3 Procedure

In this lab was several different materials observed through a NaI(Tl) scintillator with dimensions $7.62\,\text{cm} \times 7.62\,\text{cm}$ and a high purity germanium detector. The high purity germanium detector had to be cooled by liquid nitrogen since the band gaps were so small in the semiconductor. This avoided thermal excitation's which otherwise could have given of false signals [3].

The samples and the detectors were none the less placed behind lead brick as shielding and the detector was connected to a amplifier and a multi channel analyser (MCA). The signals from the scintillators and the semiconductor was then observed through the program MAESTRO- Multichannel Analyzer Emulation Software [4]. This program worked by plotting the the number of counts detected for different channels which corresponded to different energies [3].

Three pure samples(^{60}Co, ^{137}Cs and ^{22}Na) were observed through this setup with the NaI(Tl) scintillator and the high purity germanium detector. Their shapes were compared to the decay scheme for different nuclei and by this were they identified. A mystery rock was also observed with the high purity germanium detector and through the obtained energy peaks could the compound in the rock be identified. A BGO scintillator was also used for a different sample of Americium-Beryllium which was placed in a water tank. This measurement was otherwise done as the ones above [3].

From the three identified sample were five peaks with known energies used for calibrating the data in python. After this was the full width half maxima found through the formula $\text{FWHM} = 2\sqrt{2\ln(2)}\sigma$, where σ was the standard deviation of the Gaussian fit [5]. After this was the relative annihilation peak of ^{22}Na from the scintillator compared to tabulated values of this peak. This gave information about the relative peak intensity [3].

Next task was to find the internal conversion factor α for the nuclei ^{137}Ba, which correspond to the nuclei ^{137}Cs. This was found by doing a integral of the peak corresponding to the internal conversion x-ray and the full energy peak and using the known efficiencies of the Auger electrons. The third task was to find the binding energy of the deuteron. This was done with the americium-beryllium neutron source which was surrounded by water and the BGO detector. The last task was to associate the energy peaks from the background with an isotope. One was then tasked to compare the background spectrum to the mystery rock spectrum [3].

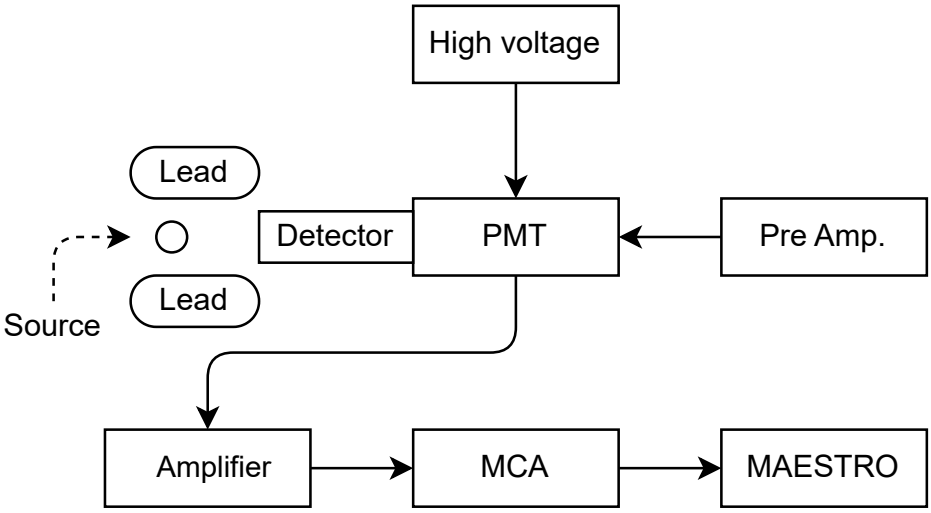


Figure 2: This figure shows the setup which was used.

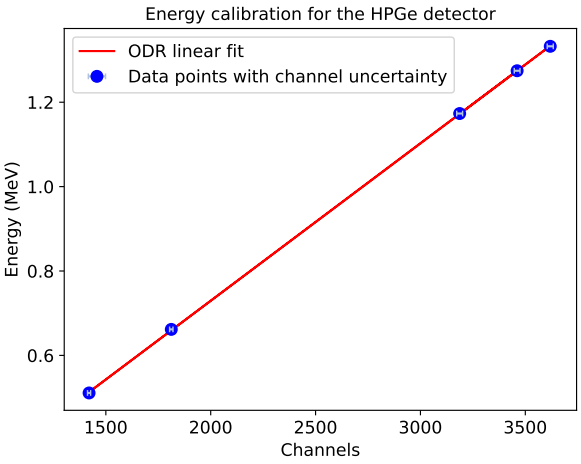
4 Result & analysis

4.1 Energy calibration of the NaI(Tl) and HPGe detectors

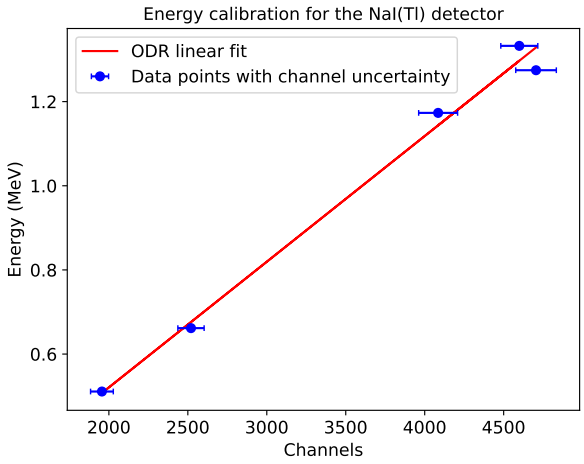
The calibration was done by performing a linear fit using the known energies in Table 1 and fitting a Gaussian on the corresponding peaks to obtain the channel number and uncertainty. The fitting parameters can be found in Table 2. The spectrum was then calibrated using these parameters so that the x-scale is in MeV in further tasks. The HPGe detector had very small errors while the errors for the NaI(Tl) had considerably larger errors. For the HPGe the fit was inside one standard deviation for all data points and for the NaI(Tl) only one data point is outside one standard deviation, though only barely.

Table 1: The known energies used for calibration of the detectors.

Spectrum:	^{22}Na	^{137}Cs	^{60}Co	^{22}Na	^{60}Co
Energy [MeV]:	0.511 00	0.661 66	1.1732	1.2746	1.3325



(a) Calibration of the High-Purity Germanium detector. The error is very small and is light blue to be visible on the blue data point markers.



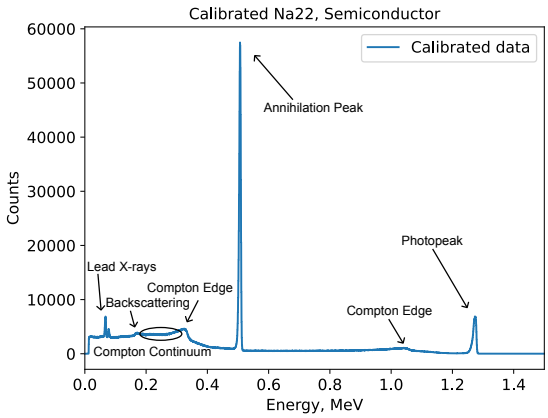
(b) Calibration of the NaI(Tl) detector.

Figure 3: Calibration of the two used detectors. The fitting parameters can be seen in Table 2.

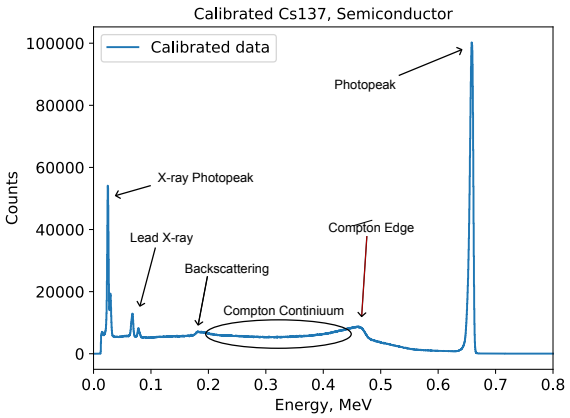
Table 2: Fitting parameters from the calibration fit. The k-value for HPGe has an error by it is not written since it is extremely small and that many significant figures cannot be justified. The fit was done as a linear $y = kx + m$ function.

	HPGe	NaI(Tl)
k	0.000 37 MeV	$(0.000\,30 \pm 0.000\,02)$ MeV
m	(-0.017 ± 0.003) MeV	(-0.077 ± 0.063) MeV

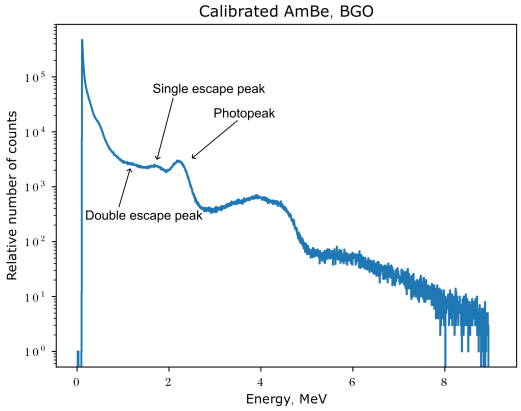
4.2 Main features of gamma-spectra



(a) Main features marked on the ^{22}Na spectrum taken with the HPGe detector.



(b) Main features marked on the ^{137}Cs spectrum taken with the HPGe detector.



(c) Main features marked on the AmBe spectrum taken with the BGO detector during the neutron lab.

Figure 4: Main features marked on different spectra taken during the gamma and neutron lab.

The photopeak (or full energy peak) is the peak with complete photoelectric absorption inside the detector. The Compton edge appears when maximal Compton scattering occurs, that is at 180 degrees. The Compton continuum appears when Compton scattering occurs at other angles and thus doesn't have the maximal energy deposition. In the ^{22}Na-spectrum it can be seen that there is a β^+ decay since there is an annihilation peak at 0.511 MeV. One can also observe lead x-rays in the spectrum which are from the gammas exciting the lead shielding which then decays and emit x-rays. The backscatter peak is from when the gamma-ray Compton scatter on the lead shielding and then interacts with the detector. The single escape peak is a peak with 0.511 MeV lower energy than the corresponding photopeak, while the double escape peak is a peak with 1.022 MeV lower energy than the corresponding photopeak. These energies correspond to one or two electrons (or positrons) escaping the detector with its mass-energy.

4.3 Full width at half maximum (FWHM) as a function of gamma-ray energy

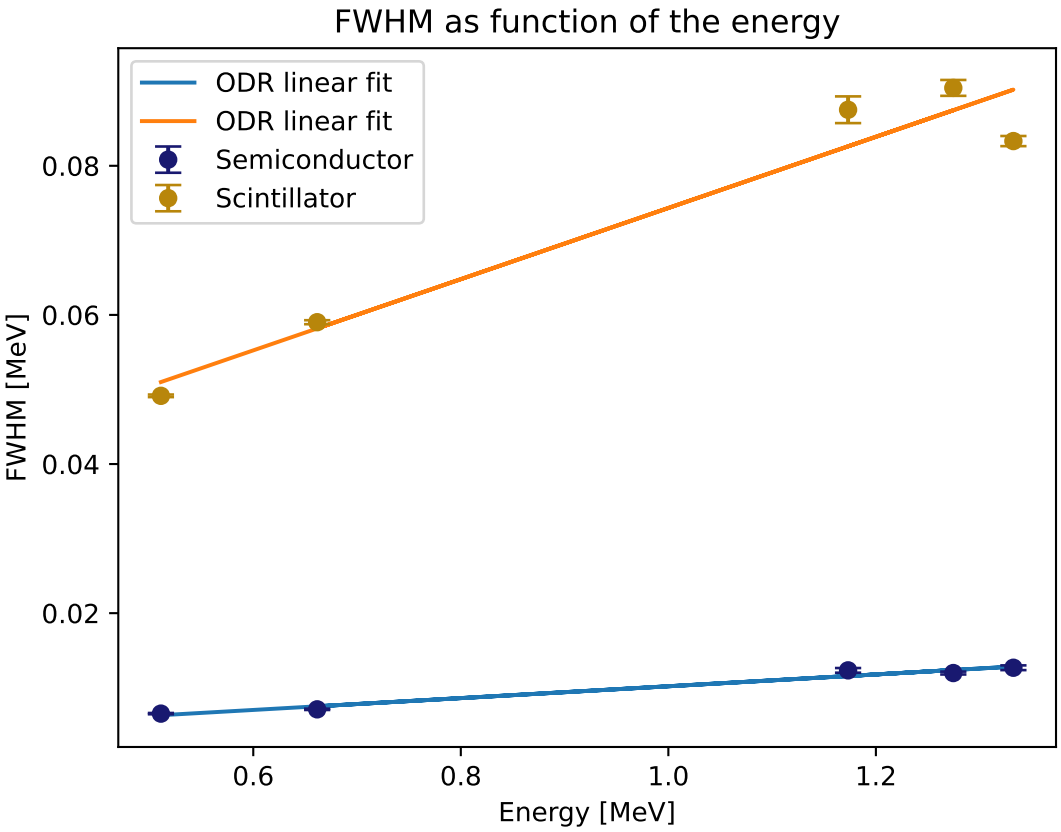


Figure 5: This figure shows the FWHM for the two detectors as well as a linear fit of the data points.

The plot shows clearly that the FWHM is increasing with increasing energy, however, it cannot be said that the increase is linear. Most data points lie multiple standard deviations from the linear fit. The scintillator detector increases more rapidly than the semiconductor detector. Also, the semiconductor detector is more linear than the scintillator.

4.4 ^{22}Na gamma-spectrum and relative peak intensity

The ratio is $R = I_0^{\text{Annihilation peak}}/I_0^{\text{Photopeak}}$ and by using Equation 7, the ratio may be written as

$$R = \frac{I_{\text{fe}}^{\text{ap}} \epsilon_{P/T}^{\text{pp}} \epsilon_{\text{eff}}^{\text{pp}}}{I_{\text{fe}}^{\text{pp}} \epsilon_{P/T}^{\text{ap}} \epsilon_{\text{eff}}^{\text{ap}}}. \quad \text{(ap is short for annihilation peak, and pp is short for photopeak)} \tag{8}$$

The solid angle is the same for both peaks so it have been ignored. From [6] the peak to total efficiency is $\epsilon_{P/T}^{\text{ap}} = 0.59$ and $\epsilon_{P/T}^{\text{pp}} = 0.33$. The intrinsic efficiency $\epsilon_{\text{eff}} = I_{\text{det}}/I_{\text{in}} = 1 - e^{-\mu x}$ where μ is the linear attenuation coefficient and $x = 7.62\,\text{cm}$, where it is assumed the particle travels on average the length of the detector. From [7] the linear attenuation coefficient is $\mu^{\text{ap}} = 0.34\,\text{cm}^{-1}$ and $\mu^{\text{pp}} = 0.18\,\text{cm}^{-1}$. Carrying out a Gaussian fit of the peaks and then using the area of the Gaussian, the full energy peaks are (1234 ± 6) Counts for the annihilation peak and (311 ± 5) Counts for the photopeak. The resulting ratio after calculations is $R = (1.79 \pm 0.03)$. The expected value is 1.8 so the experimental value is 0.3σ from the theoretical value. The theoretical value is 1.8 since 90 % decays via β^+ which then annihilates to

two gammas, while 10 % decay with EC and are not detected. The new state then decays with a gamma. Thus all decays lead to one gamma, while 90 % decay with two gammas, thus $0.9 \times 2 \times 1 = 1.8$.

4.5 ^{137}Cs and the internal conversion coefficient of ^{137}Ba

Using Equation 5 in combination with Equation 6 and replacing the intensities with Equation 7 and cancelling the solid angle since it is the same for all particles, results in the equation

$$\alpha_k = \frac{I_{\text{fe}}^{\text{x-ray}} \epsilon_{\text{eff}}^{\gamma} \epsilon_{P/T}^{\gamma}}{I_{\text{fe}}^{\gamma} \epsilon_{\text{eff}}^{\text{x-ray}} \epsilon_{P/T}^{\text{x-ray}} \omega_{\text{k}}}. \tag{9}$$

From the lab manual it is known that $\epsilon_{\text{eff}}^{\gamma} \epsilon_{P/T}^{\gamma} = 0.3$ and $\epsilon_{\text{eff}}^{\text{x-ray}} \epsilon_{P/T}^{\text{x-ray}} = 0.9$. From [6] the K-fluorescence yield $\omega_k = 0.901$, thus the value is only corrected for the K-shell. The full energy intensities are the area of the peaks. For the x-ray there is two peaks where the areas where added, this however results in a bit too small of an area due to how the Gaussian fit was performed. The resulting area was (754 ± 12) Counts for the gamma peak and (163 ± 12) Counts for the x-ray peaks after correcting for background. This gives an internal conversion coefficient of $\alpha_k = (0.080 \pm 0.006)$. Using [8] and the measured wavelength of the gamma $\lambda = (658 \pm 3)$ keV and multipolarity M4, the literature value is $\alpha_k = (0.093 \pm 0.002)$. That is the experimental value is 2.2σ from the literature value.

4.6 Binding energy of the deuteron

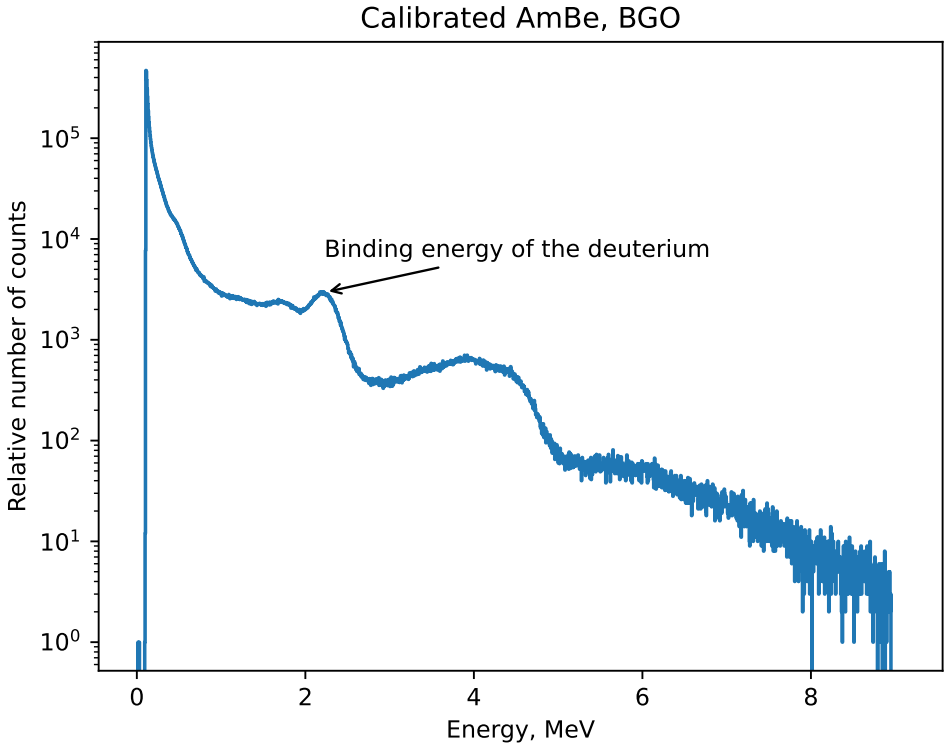


Figure 6: This figure shows the binding energy of the deuterium.

The deuteron is created in the experiment by hydrogen in water absorbing a neutron and producing a γ with the corresponding energy.By fitting a Gaussian function to Figure 6 was the binding energy of the deuterium found to be (2.2 ± 0.2) MeV. The theoretical value is $(2.224\,575 \pm 0.000\,009)$ MeV according to [9]. Thus, the measured value in the lab is very reasonable having the theoretical value well inside one standard deviation.

4.7 Background radiation

The mystery rock was during the lab identified to be ^{238}U, which can be seen in Appendix A. In Figure 7 can one see the data of the background data overlayed the data of ^{238}U. The peaks of the background data

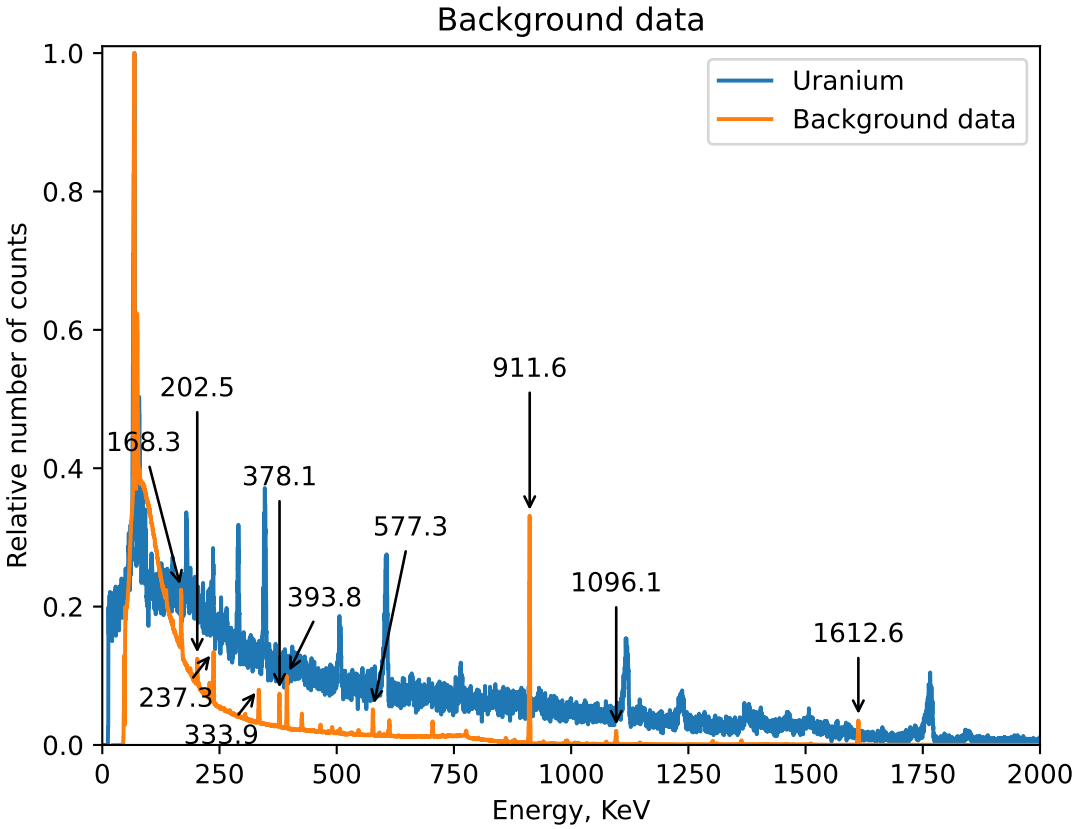


Figure 7: This figure shows the data of the ^{238}U and the background. Both are normalised according to their respectively largest peak. Peaks in the background data are shown with arrows.

One can in Figure 7 see that the data from the background measurement and the data from the ^{238}U do not overlap. In Table 3 can one see that the radiation mostly comes from ^{232}Th.

Table 3: This table shows the origin and nuclei of the data shown in Figure 7

.

Experimental energy [keV]	Similar tabulated energy [keV]	Nuclei	Origin
(168.3 ± 2.8)	166.0	^{88}Kr	Fission
(202.5 ± 2.8)	205.3	^{235}U	Natural
(237.3 ± 2.8)	238.6	^{212}Pb	^{232}Th
(333.9 ± 2.8)	338.4	^{228}Ac	^{232}Th
(378.0 ± 2.8)	380.4	^{125}Sb	Fallout
(393.8 ± 2.8)	396.6	^{138}Xe	Fission
(577.3 ± 2.8)	583.0	^{208}Tl	^{232}Th
(911.6 ± 2.9)	911.2	^{228}Ac	^{232}Th
(1096.1 ± 3.0)	1099.3	^{59}Fe	Fallout
(1612.6 ± 3.2)	1620.7	^{212}Bi	^{232}Th

5 Conclusion

From the calibrations plot in Figure 3 can one see that the calibration of the semiconductor has much smaller error bars than the scintillator. This is because the energy differences of the different measurements of the semiconductor are much smaller than that of the scintillator. The reason why the semiconductor is much narrower is because the band gaps are a lot smaller than the scintillator. This implies that the germanium detector has a lot better resolution and a higher efficiency. So i one wants to detect very specific energies, a Germanium detector is a much better choice. However, due to the price and the difficulties of creating germanium semiconductors is not a suitable choice when one wants to measure gamma rays over a large area. A scintillator is much cheaper and easier to make into larger slabs, thus would this be a better choice for large-area detection.

Annihilation peaks and escape peaks may seem similar but are two very different phenomena. The Annihilation peak is due to a positron and electron annihilating inside the source and sending out two photons with 511 keV each and depending on detector geometry, one or two are detected. The escape peak, however, is due to pair production inside the detector creating an electron-positron pair, then an annihilation event occurs and sends out two photons. However, one or both might leave the detector undetected, thus removing 511 keV or 1022 keV from the detection. So the main difference is if the annihilation is occurring in the source or in the detector and if the peak is at 511 keV or 511 keV or 1022 keV lower than the photopeak.

In Table 4 can one see that the table mostly shows that the energies correspond to the decay chain of ^{238}U. This is very logical since ^{238}U is a fairly common type of mineral. The reason why we see other origins such as "Natural" and "Fission" form ^{137}Cs may come from the reason that there a several close energies for these values. The natural energy may also just come from natural sources inside the rock since ^{238}U often contain small amount of ^{235}U. The fission from ^{137}Cs may come from small particles left from this from earlier testing on the same detector. However this might not be the case and this energy properly just arises from similar tabulated energies.

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A Mystery rock

Regarding the analysis of the mystery rock one can see its spectra in Figure 8. Comparing the peaks in Figure 8 to the values inTable 4 one can clearly see that almost all peaks can be traced to ^{238}U , thus the conclusion that the mystery rock is made up of radioactive ^{238}U can be made.

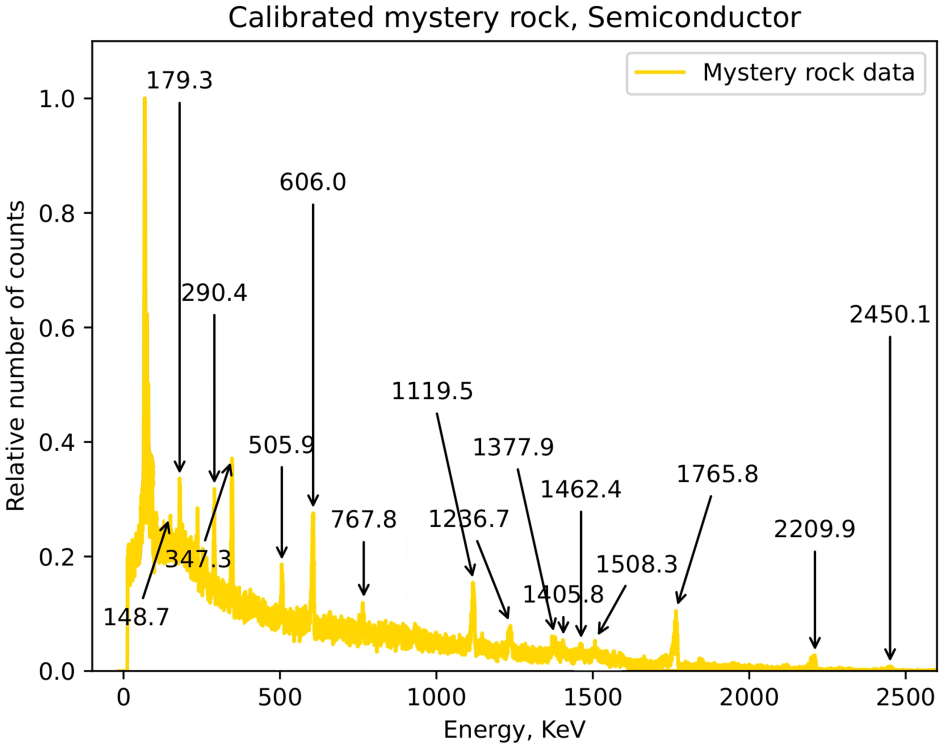


Figure 8: This data shows the data from the mystery rock, normalised to its own data.

Table 4: The energies from Figure 8 were compared to the most similar tabulated values [10].

Experimental energy [keV]	Similar tabulated energy [keV]	Nuclei	Origin
(148.7 ± 2.8)	143.8	^{235}U	Natural
(179.3 ± 2.8)	176.6	^{137}Cs	Fission
(290.4 ± 2.8)	295.1	^{214}Pb	^{238}U
(347.3 ± 2.8)	352.0	^{214}Pb	^{238}U
(505.9 ± 2.8)	510.6	^{208}Ti	^{232}Th
(606.0 ± 2.8)	609.3	^{214}Bi	^{238}U
(767.8 ± 2.9)	768.4	^{214}Bi	^{238}U
(1119.5 ± 3.0)	1120.3	^{214}Bi	^{238}U
(1236.7 ± 3.0)	1238.1	^{214}Bi	^{238}U
(1377.9 ± 3.1)	1377.7	^{214}Bi	^{238}U
(1405.8 ± 3.1)	1408.0	^{214}Bi	^{238}U
(1462.4 ± 3.1)	1462.0	^{228}Ac	^{232}Th
(1508.3 ± 3.1)	1509.2	^{214}Bi	^{238}U
(1765.8 ± 3.2)	1764.5	^{214}Bi	^{238}U
(2209.9 ± 3.5)	2204.1	^{214}Bi	^{238}U
(2450.1 ± 3.6)	2447.7	^{214}Bi	^{238}U